

MULTIPHASE PWM CONTROL STRATEGY FOR PMSM WITH SEGMENTED STATOR WINDINGS

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OBJECTIVES

- To implement a multiphase converter with PWM strategy for control of Permanent Magnet Synchronous Motor(PMSM).
- To operate a PMSM with segmented stator windings as an electronic geared motor.

INTRODUCTION

SEGMENTED MACHINES

- Segmented motors are motors with modular segmented stator windings which are designed and manufactured with identical segments.
- Can share a flexible production line for a wide variety of applications.

- Segmented electrical machines are highly flexible.
- They can be manufactured at relatively low costs.
- By the concept of segmentation, manufacturing automation of small motors is possible at low costs.
- Easier to manufacture motors with high no.of pole pairs.

Using PWM control scheme for a Motor with multi-phase open-end winding topology we can:

- Achieve independent PWM control for each winding.
- Provide flexibility for converters to control the various windings as an independent load.
- Using this technique, a five-phase PMSM with open-end windings can be controlled using five H-bridge VSIs with a common DC bus.

The advantages of this method are:

- Reduced switching losses.
- Maximized drive performance.
- Achieve wider speed and torque control range with same DC-link voltage.
- Improve efficiency over a wide speed control range.

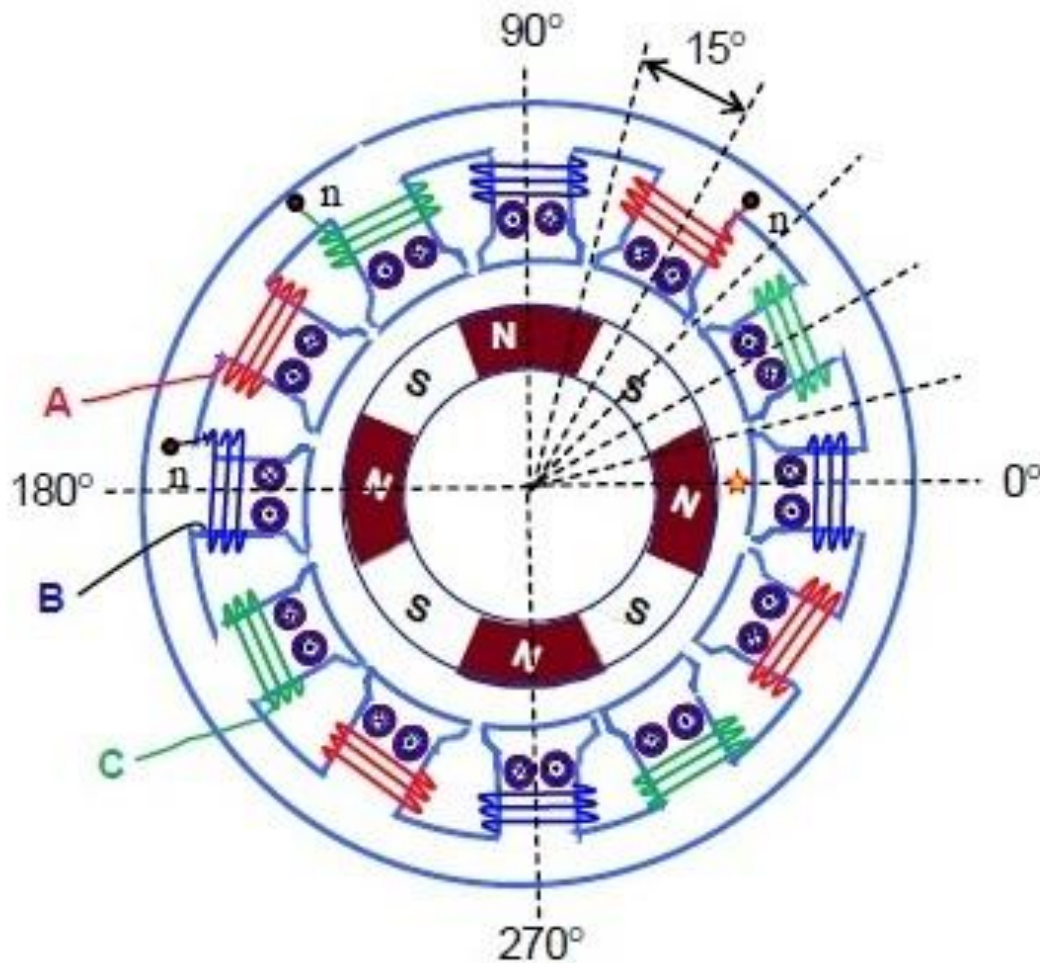
- Control each winding independently while they are functionally configured either in series or parallel.
- When stator windings in same phase are connected in series, motor back emf constant, K_b will increase.
- If they are connected in parallel, motor torque constant, K_t will increase.

SWITCHED STATOR WINDING METHOD

- By adjusting connections of stator windings with open-ends, we can change the torque-speed chara. of PMSM.
- This mechanical connection must be done before application and cant be changed during operation.
- Parallel connection will cause current unbalancing problems.

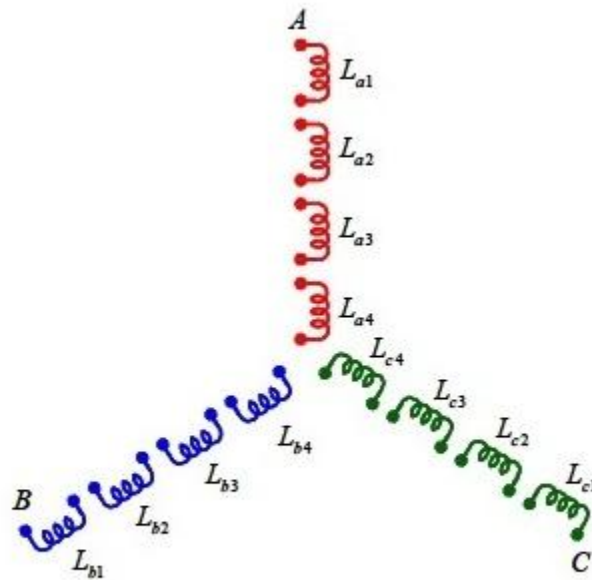
MULTI PHASE CONVERTER FOR SEGMENTED PMSM

- Multi-phase inverter can control of each segmented windings to achieve accurate current sharing under parameter uncertainties.
- PWM technique is applied for control of multi-phase inverter for the segmented motor.

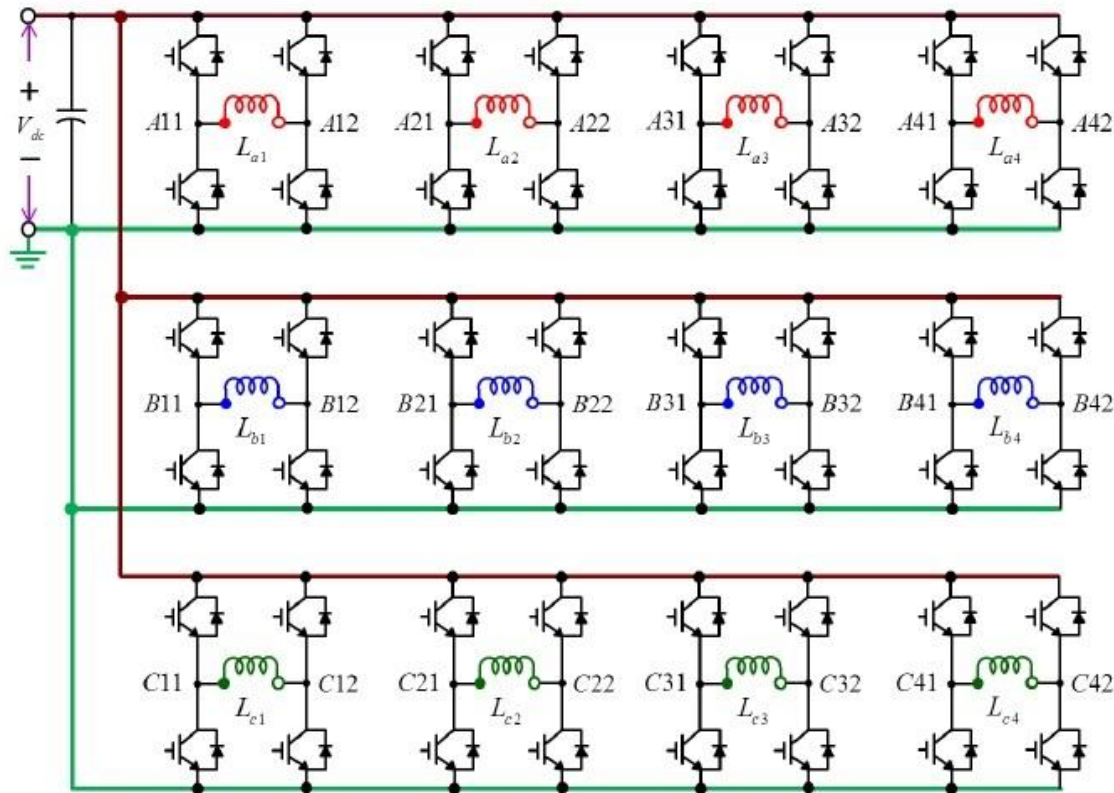


Cut-view of multiphase segmented PM synchronous motor.

- Each phase has four independent segmented coils which are open-end winding as shown in the below figure.



segmented stator with open-end windings



Proposed multiphase converter topology.

This is the multi-phase converter topology for the proposed mutli-phase PWM control scheme.

- Four segments are present for each phase of three-phase segmented PMSM.
- 12 windings need to be controlled.
- 12 H-bridge converters are present to control voltage applied to each winding .
- Multi-phase converter requires high number of power semiconductor devices.

- Each segment winding is viewed as a controllable stator magnetic field.
- Vector summation of segmented fields will produce a composite stator magnetic field.
- Stator and rotor magnetic fields interact to produce a controllable electric torque.
- A phase winding with N multiple segments provides $(2N-1)$ degree-of-freedom for control of stator field of corresponding phase.

Benefits Of Multi-phase Converters:

- Each segment contributes to the generated torque
- Voltage applied to motor is shared by more power devices.
- Distributed power dissipation.
- Losses are distributed in N H-bridge power modules.
- Losses for each module is relatively small.

- Low rating of power devices can be employed.
- Current ratings of these power devices are kept the same.
- DC-link voltage can be reduced to

$$V_{dc} = \frac{V_{dc,3W}}{N}$$

N- Number of segments per phase
 $V_{dc,3W}$ - DC-link voltage required

MULTIPHASE PWM CONTROL STRATEGY

The PWM control for the multiphase converter can:

- Reduce torque ripples by
 - Phase Shedding Control
 - Phase Shifting Control
- Results in overall improvement of speed-torque control range, efficiency.

PHASE SHEDDING CONTROL:

- By using this method, we can reduce torque ripples in multiphase converters.
- It improves the light-load efficiency by turning phases OFF at light loads.
- In doing so the power consumption of various circuit components is saved for every phase that is disabled.

- The developed electric torque of the segmented motor is a summation of torques produced by excited windings.

$$T_e = \sum_{i=1}^N \tau_{ai} + \tau_{bi} + \tau_{ci}$$

- When more segmented windings are activated, higher voltage and current can be applied to the motor.

- The maximum rated power is limited by the motor due to its thermal dissipation capability.
- The maximum torque limit is usually limited by the available maximum phase current.
- The disadvantage of this approach is that it cannot be extended to high voltages.

PHASE SHIFT CONTROL:

- When multiple PWM converters are connected in parallel, it is advantageous to operate all converters at the same frequency but with certain phase shift among their PWM signals.
- Provides major advantages for current sharing and current ripple reduction.
- Output torque and input current ripples can be reduced

CONCLUSION

- A multiphase PWM modulation strategy is developed for the control of the segmented PMSM to achieve a various torque-speed performance with a lower dc-link voltage at the cost of an increase of switching devices with lower voltage ratings.
- The PWM strategy provides advantages of ripple current reduction and conduction losses reduction.
- Phase shedding technique can be used to improve system efficiency under light load operating conditions.

REFERENCES

- [1] Chiao-Chien Lin and Ying-Yu Tzou, “An Innovative Multiphase PWM Control Strategy for a PMSM with Segmented Stator Windings,” *IEEE Applied Power Electronics Conference and Exposition*, Charlotte, NC, pp 270 - 275, 19 March 2015

THANK YOU